RAG-Induced Failures in Multi-Agent Large Language Model Debate

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***Abstract*— Multi-agent Large Language Model (LLM) systems based on collaborative debate have been proposed to improve reasoning accuracy and robustness by aggregating diverse reasoning trajectories. However, most debate frameworks implicitly assume that all participating agents behave honestly. This paper demonstrates that this assumption fails under a realistic threat model in which a single malicious agent manipulates Retrieval-Augmented Generation (RAG) to selectively retrieve externally sourced evidence that plausibly supports an incorrect conclusion. Unlike prior work that relies on prompt injection or complex persuasion pipelines, the proposed attack requires only retrieval control and uses legitimate documents whose relevance is strategically overstated or misinterpreted. Experiments on TruthfulQA, MMLU, MedMCQA, and SCALR show that RAG-based attacks can reduce majority-vote accuracy by up to 38% while increasing group consensus around false answers. These results indicate that retrieval—often used to improve factual grounding—can instead amplify misinformation when adversarially controlled, motivating stronger verification and trust calibration in multi-agent LLM systems.**

Keywords— Multi-Agent LLMs, Retrieval-Augmented Generation, Adversarial Debate, Collaborative Reasoning, LLM Safety

I. Introduction

Recent advances in Large Language Models (LLMs) have enabled their deployment as autonomous agents capable of reasoning, interaction, and cooperation. Rather than operating in isolation, modern LLM systems increasingly adopt multi-agent collaboration, where several models deliberate through structured debate, consensus mechanisms, or majority voting. Prior studies report improvements in reasoning accuracy, robustness, and coverage when multiple agents jointly solve complex tasks.

Despite these benefits, collaborative LLM systems rely on an implicit and rarely questioned assumption: all agents are honest participants. In realistic deployments—such as decision support, legal analysis, or medical reasoning—this assumption may not hold. An adversarial or compromised agent may intentionally manipulate deliberation to steer the group toward incorrect conclusions.

Existing work on adversarial threats to LLMs has primarily focused on prompt injection, data poisoning, or input perturbations. In contrast, the vulnerability of LLM-to-LLM interaction itself remains underexplored. In particular, the role of retrieval-augmented generation (RAG) in amplifying adversarial influence has not been systematically studied.

This paper addresses this gap by showing that RAG alone is sufficient to destabilize multi-agent debate. By selectively retrieving external documents and presenting them as evidence, a malicious agent can appear credible and authoritative while consistently promoting false conclusions[1]. Unlike complex persuasion pipelines, our attack relies solely on retrieval control—making it both realistic and difficult to detect[2].

II. Related work

Collaborative reasoning with multiple Large Language Models has been shown to improve accuracy and robustness compared to single-model inference. Debate-based frameworks enable agents to iteratively critique and refine responses, often leading to stronger performance on complex reasoning benchmarks [3], [4], [5], [6]. Systems such as AutoGen, CAMEL, AgentVerse, and MetaGPT formalize these interaction patterns and demonstrate their effectiveness in structured reasoning and task decomposition [7], [8], [9], [10]. However, these approaches fundamentally assume that all agents act cooperatively and share aligned objectives. As shown by the degradation trends in our experimental results, this assumption becomes a critical weakness when even a single agent behaves adversarially.

Recent studies indicate that LLMs possess strong persuasive capabilities, generating arguments that are coherent and confident regardless of factual correctness[11], [12]. Empirical analyses further show that both humans and models can be influenced by arguments that emphasize plausibility and rhetorical structure over verifiable truth[13], [14]. In the context

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of LLM-to-LLM interaction, Khan et al. [15]demonstrate that more persuasive models tend to dominate debate outcomes, a finding that aligns with the increasing adversarial agreement observed across debate rounds in our results. Initial mitigation attempts based on prompt-level warnings or alignment constraints have shown limited success [12], reinforcing the need to examine deeper structural vulnerabilities.

Most adversarial research on LLMs focuses on input manipulation, such as prompt injection, poisoning, or evasion attacks[16]. While effective in user-facing settings, these approaches do not capture interaction-level threats arising from autonomous agent coordination. Recent robustness studies emphasize that vulnerabilities may emerge from system design choices rather than model parameters alone [17]. Our experimental findings support this view, showing that debate protocols themselves can amplify adversarial influence rather than suppress it.

Retrieval-Augmented Generation (RAG) is widely adopted to improve factual grounding and reduce hallucinations by conditioning generation on external knowledge sources [18], [19], [20], [21]. However, recent evidence suggests that LLMs may over-trust retrieved content, particularly when it appears authoritative or domain-specific. While retrieval quality has been studied extensively, prior work does not examine adversarial control of retrieval in multi-agent debate settings. The accuracy drops and agreement shifts reported in our Results section demonstrate that selectively retrieved evidence can significantly distort collective reasoning, even when all retrieved content originates from legitimate sources.

Existing literature treats debate, persuasion, adversarial attacks, and retrieval augmentation largely in isolation. Our work connects these strands by showing that RAG alone constitutes a powerful adversarial vector in collaborative LLM systems. Unlike prior approaches that rely on complex persuasion pipelines, our results reveal that selective retrieval is sufficient to reduce system accuracy and induce false consensus. This highlights a fundamental gap in current multi-agent LLM designs: trust in retrieved evidence is implicitly assumed, yet never verified.

III. Methodology

We consider a standard multi-agent debate framework in which multiple LLM agents are tasked with solving a shared reasoning problem through iterative interaction. The system consists of a group of cooperative agents and a single adversarial agent that participates under the same interface and constraints. All agents have access to identical model architectures and generation capabilities, and all follow the same debate protocol. The adversarial agent does not possess privileged access to system internals, ground-truth labels, or prompt templates. Instead, its sole advantage lies in its control over retrieval queries used during Retrieval-Augmented Generation. This makes the threat model realistic and difficult to detect, as the adversary behaves indistinguishably from cooperative agents at the protocol level.

The goal of the adversary is twofold: (i) reduce the overall accuracy of the system's final decision, and (ii) increase alignment of cooperative agents with an incorrect answer. Importantly, the adversary does not aim to dominate the conversation overtly, but rather to subtly steer consensus through seemingly well-supported arguments grounded in retrieved external content. Fig 1. Flowchart of the proposed RAG-based adversarial attack in multi-agent LLM debate. A single adversarial agent, operating under identical model and protocol constraints as cooperative agents, manipulates the debate solely by controlling retrieval queries during Retrieval-Augmented Generation. By selectively introducing seemingly credible external evidence, the adversary subtly steers group consensus toward an incorrect answer, leading to reduced majority-vote accuracy and increased adversarial alignment.

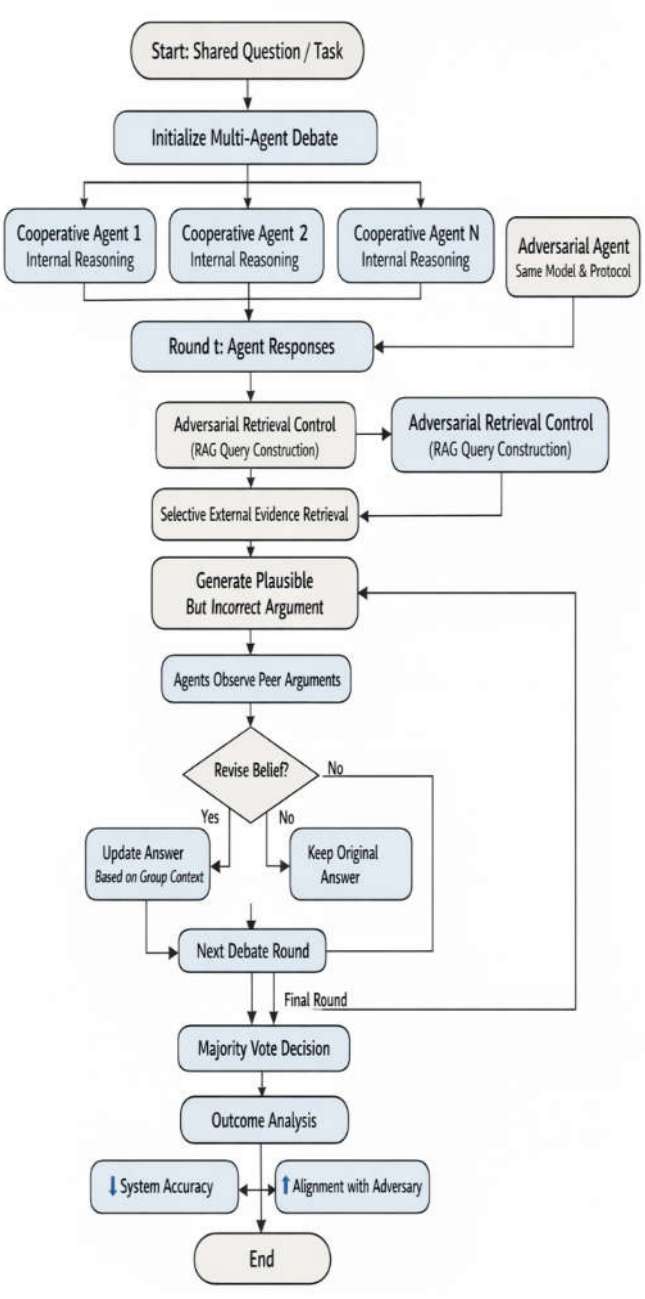


Fig. 1. Flowchart of the proposed RAG-based adversarial attack in multi-agent LLM debate.

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A. Multi-Agent Debate Protocol

The collaborative debate proceeds in discrete rounds. Initially, each agent independently generates an answer to the shared question based solely on its internal reasoning. In subsequent rounds, agents observe the responses of their peers and may revise their answers accordingly. This process is repeated for a fixed number of rounds, after which a final decision is produced using majority voting.

This debate protocol is intentionally simple and mirrors commonly used multi-agent reasoning frameworks. No explicit trust scoring, verification, or weighting is applied to agent responses, reflecting realistic deployments where agents are assumed to be equally reliable.

B. RAG-Only Adversarial Strategy

Unlike prior adversarial approaches that rely on elaborate persuasion pipelines [22],[23], our methodology employs a single, minimal attack mechanism: retrieval manipulation. At each debate round, the adversarial agent constructs a retrieval query using the original question, its chosen incorrect answer, and a summary of other agents' responses. This query is submitted to an external knowledge source such as Wikipedia or domain-specific corpora. From the retrieved passages, the adversarial agent selectively retains content that appears authoritative, ambiguous, or weakly related to the task, yet can be framed to support the incorrect conclusion. Crucially, retrieved passages are not fabricated; they are real texts whose relevance is strategically overstated or misinterpreted. The adversarial response is then generated by grounding its argument explicitly in this retrieved content, increasing perceived credibility. By repeatedly injecting such retrieval-grounded arguments across debate rounds, the adversarial agent gradually shifts the belief state of cooperative agents. This demonstrates that selective evidence exposure alone is sufficient to manipulate collective reasoning. Equations (1)–(5) formalize the proposed RAG-only adversarial strategy, showing that manipulation arises solely from selective retrieval and grounding, without prompt modification, model asymmetry, or explicit persuasion heuristics. Let q denote the original question and $A^{(t)} = \{a_1^{(t)}, \dots, a_N^{(t)}\}$ the set of agent responses at debate round t. The adversarial agent selects an incorrect target answer y^-. At each round t, the adversarial retrieval query is constructed as (1).

$$r^{(t)} = \phi\left(q, y^-, A^{(t-1)}\right) \tag{1}$$

where $\phi(\cdot)$ is a query construction function that combines the original question, the adversarial answer, and a summary of peer responses.

Given an external knowledge corpus D, the retrieval module returns a ranked set of passages in (2).

$$P^{(t)} = Retrieve\left(r^{(t)}, D\right) \tag{2}$$

From $P^{(t)}$, the adversarial agent applies a selective filtering function ψ(·) to retain passages that are rhetorically supportive of y^- in (3).

$$\tilde{P}^{(t)} = \psi\left(P^{(t)} \mid y^-\right) \tag{3}$$

The adversarial response is then generated by conditioning on the selected passages in (4).

$$a_{adv}^{(t)} \sim P_\theta\left(a \mid q, \tilde{P}^{(t)}\right) \tag{4}$$

Finally, the influence of the adversarial agent on cooperative agents can be modeled as a shift in their belief distribution over candidate answers in (5).

$$P^{(t+1)}(y) = (1-\alpha)\,P^{(t)}(y) + \alpha\, I\left[y = y^-\right] \tag{5}$$

where $\alpha \in (0,1)$ represents the persuasion strength induced by retrieval-grounded arguments and $I[\cdot]$ is the indicator function.

IV. Experimental Setup

A. Datasets

To evaluate the generality of the proposed attack, we conduct experiments across four benchmarks covering diverse reasoning domains. TruthfulQA focuses on susceptibility to plausible misinformation, making it particularly suitable for studying persuasive falsehoods. Massive Multitask Language Understanding (MMLU) provides a broad academic evaluation across numerous subjects, enabling assessment of general reasoning robustness. Medical Multiple-Choice Question Answering (MedMCQA) represents high-stakes medical decision-making, where erroneous conclusions can be especially harmful. Statutory Case Law Reasoning (SCALR) evaluates legal reasoning, a domain inherently sensitive to selective evidence and precedent interpretation. Together, these datasets allow us to examine both general-purpose and domain-specific vulnerabilities of collaborative LLM systems under adversarial retrieval influence. The characteristics and statistics of the evaluation datasets used in this study are summarized in Table I.

B. Evaluation Metrics

We adopt two complementary metrics to quantify adversarial impact. The first, ΔAccuracy, measures the change in final majority-vote accuracy relative to a no-attack baseline. This captures system-level degradation.

Let:

- A_{base} = baseline accuracy without adversarial intervention
- A_{adv}= accuracy after adversarial debate

TABLE I. Description and Statistics of Evaluation Datasets

Dataset	Domain / Focus	# Questions	Format
MMLU	General knowledge across 57 academic subjects (STEM, humanities, social sciences, law, medicine)	14,042	Multiple-choice
TruthfulQA	Factual accuracy and misinformation susceptibility	817	Open-ended / multiple-choice
MedMCQA	Medical decision-making (diagnosis, treatment, pharmacology, pathology)	>194,000	Multiple-choice
SCALR	Legal reasoning, statutory interpretation, case analysis	15.254	Text-based / structured QA

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The metric is defined as (6):

$$\Delta Accuracy = A_{\text{adv}} - A_{base} \quad (6)$$

An effective adversarial attack gives: $\Delta Accuracy < 0$. The bigger the negative value, the greater the disruption of collaborative thinking, which is the worldwide effect of the attack.

The second, ΔAgreement, measures the increase in the number of cooperative agents that adopt the adversarial answer by the final debate round. This captures agent-level persuasion dynamics.

Let:

- $Agree_0$ = number of the agents whose first response satisfies the adversarial response.
- $Agree_T$ = respondents who agree with the adversarial answer at the end of the last round T.

Then:

$$\Delta Agreement = Agree_T - Agree_0 \quad (7)$$

A positive increase indicates: $\Delta Agreement > 0$. This reflects successful persuasion, capturing the local behavioral influence of the adversarial agent.

An attack is considered successful when it simultaneously reduces system accuracy and increases alignment with the adversarial agent, providing a holistic assessment of manipulation effectiveness.

V. Results

Fig 2 (Accuracy Degradation under RAG Attack) illustrates the impact of the proposed RAG-based adversarial strategy on model accuracy across all datasets and models. The figure reports ΔAccuracy (Attack − Clean), where negative values indicate performance degradation. A consistent and substantial accuracy drop is observed for all evaluated models and benchmarks, confirming that the attack generalizes beyond a specific architecture or domain. Notably, GPT-3.5 exhibits the most pronounced degradation on MMLU and TruthfulQA, while even stronger models such as GPT-4o experience non-negligible losses. This result highlights that retrieval-based manipulation remains effective even against higher-capacity models, suggesting that model scale alone is insufficient as a defense when retrieval channels are adversarially controlled.

Fig 3 (Adversary Agreement across Debate Rounds) provides insight into the underlying mechanism driving the observed accuracy collapse. Across all datasets, adversary agreement increases monotonically over successive debate rounds, indicating that cooperative agents progressively align with the adversarial agent's incorrect position. This trend demonstrates that the attack does not merely introduce random noise or confusion; instead, it systematically steers collective reasoning through selectively retrieved evidence. Importantly, agreement growth coincides with accuracy degradation, reinforcing the conclusion that the attack succeeds by reshaping group consensus rather than by isolated prediction failures.

Table II (Attack Effectiveness Summary) quantitatively consolidates these observations by jointly reporting accuracy degradation and adversary influence. For all datasets, group accuracy decreases from the first to the final round, while adversary agreement increases substantially. The Attack Ratio further reveals differences in attack efficiency across tasks, with MMLU exhibiting the highest ratio—indicating the largest accuracy loss per unit increase in adversarial agreement. This finding suggests that broad-knowledge benchmarks are particularly vulnerable to retrieval-based misinformation, likely due to the greater availability of plausible but misleading external evidence.

Fig 4 illustrates the evolution of group-average accuracy of the non-adversarial GPT-3.5 agents across successive debate rounds under adversarial influence. A consistent and monotonic decline in accuracy is observed across all four datasets—TruthfulQA, MMLU, MedMCQA, and Scalr—indicating that iterative interaction does not improve collective reasoning when a malicious agent is present. Notably, the degradation is most pronounced between the first and second rounds, suggesting that early exposure to adversarial arguments has a disproportionately strong impact on cooperative agents. Rather than correcting erroneous reasoning through mutual critique, the debate process amplifies misleading information introduced by the adversary. This effect persists across both general-domain benchmarks (TruthfulQA, MMLU) and domain-specific tasks (MedMCQA, Scalr), highlighting the dataset-agnostic nature of the vulnerability.

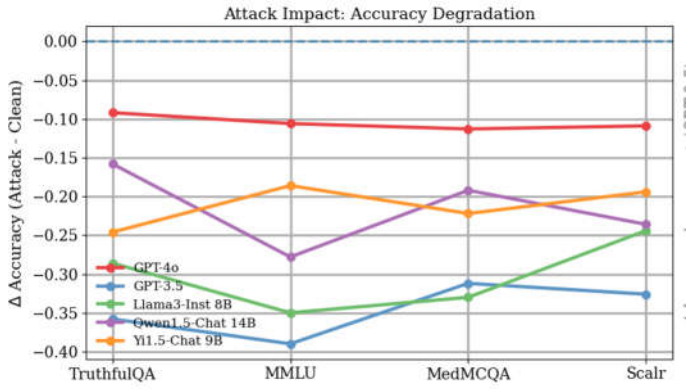


Fig. 2. Accuracy degradation of collaborative LLM debate under RAG-only adversarial attack.

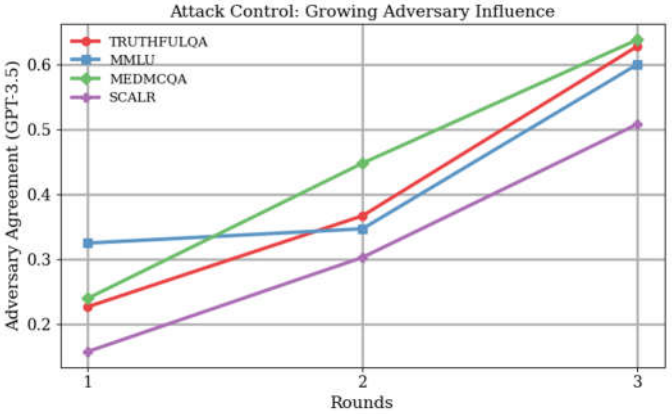


Fig. 3. Growth of adversarial agreement across debate rounds.

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TABLE II. SUMMARY OF RAG-BASED ADVERSARIAL ATTACK EFFECTIVENESS ACROSS DATASETS

Dataset	Group Acc (R1)	Group Acc (R3)	Δ Accuracy	Adv Agr (R1)	Adv Agr (R3)	Δ Agreement	Attack Ratio
TRUTHFULQA	0.555	0.283	**−0.272**	0.227	0.628	**+0.401**	0.678
MMLU	0.616	0.335	**−0.281**	0.325	0.600	**+0.275**	1.022
MEDMCQA	0.536	0.306	**−0.230**	0.240	0.638	**+0.398**	0.578
SCALR	0.533	0.323	**−0.210**	0.158	0.508	**+0.350**	0.600

Overall, the combined evidence from Fig 2, Fig 4 and Table I and II demonstrates that the proposed RAG-based attack is both effective and systematic: it consistently reduces collective accuracy while simultaneously increasing adversarial control over group decisions. These results confirm that retrieval mechanisms, when adversarially, can undermine the core assumptions of collaborative LLM debate and transform collective reasoning from a robustness-enhancing strategy into a vulnerability[24], [25]. The results reveal a fundamental tension in multi-agent LLM systems: mechanisms designed to improve reasoning—such as debate and retrieval—can be repurposed to amplify adversarial influence. Retrieval-augmented generation introduces an implicit trust channel, where externally sourced content is assumed to be factual. When this channel is adversarially controlled, misinformation gains disproportionate persuasive power. These findings suggest that robustness in collaborative AI cannot be achieved solely through diversity or interaction. Instead, explicit verification, cross-agent evidence auditing, and trust calibration mechanisms are required to prevent manipulation.

VI. LIMITATIONS AND FUTURE WORK

This study has several limitations. First, experiments are computationally intensive, limiting the exploration of very large agent populations[26]. Second, retrieval sources are assumed to be static and unverified, whereas real-world systems may employ dynamic or curated corpora[27]. Third, we focus exclusively on text-based debate protocols; multimodal or tool-augmented agents may exhibit different dynamics[28], [29]. Despite these limitations, the observed vulnerabilities are structural and likely to persist across settings[30], [31]. Future research will address these limitations along several directions. First, scalable experimentation with larger agent populations will be enabled through approximate coordination strategies, agent sampling, or hierarchical debate structures, allowing the study of persuasion dynamics at scale with reduced computational cost. Second, future work will consider dynamic and curated retrieval pipelines, including trust-aware retrieval, source verification, and adaptive corpus updating, to better reflect real-world deployment scenarios. Third, the framework will be extended beyond text-only interactions to include multimodal agents and tool-augmented reasoning, such as agents with access to external calculators, symbolic solvers, or visual inputs, to examine whether persuasion-driven vulnerabilities persist under richer interaction modalities. Finally, future studies will explore defensive mechanisms, including persuasion-aware aggregation rules, adversarial agent detection, and robustness-aware debate protocols, with the goal of mitigating the structural weaknesses identified in collaborative LLM systems.

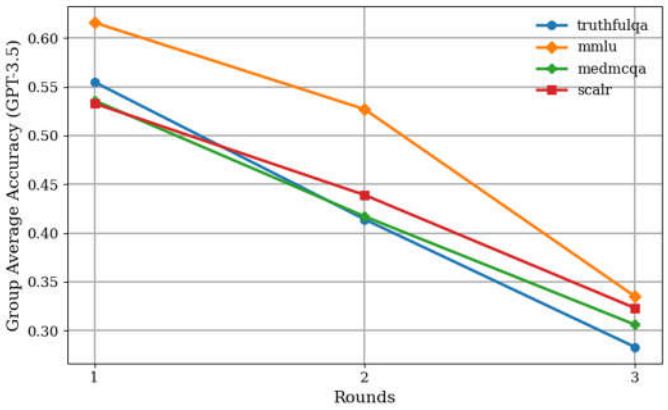


Fig. 4. Evolution of group-average accuracy across debate rounds under adversarial influence.

VII. CONCLUSION

This paper demonstrates that collaborative LLM debate systems are inherently vulnerable to RAG-only adversarial attacks**.** By selectively controlling retrieval queries[32], a single malicious agent can manipulate group reasoning, reduce accuracy, and induce consensus around false conclusions—without altering prompts, models, or protocols. These findings underscore the need for adversarial-aware retrieval design and principled coordination mechanisms in multi-agent AI systems. As LLM agents become increasingly autonomous and interconnected, addressing retrieval-based manipulation will be critical to ensuring reliable and safe deployment.

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